



EmpowerWork
Opportunity for Every Ability.

Problem



Solution



EmpowerWork
Opportunity for Every Ability.

***MOHAMED GAMAL
IS***

***KHALED WALEED
IS***

***MOHAMED HASSAN
IS***

***RAWAN MOHAMED
IS***

***NADEEN EHAB
CyS***

***MAZEN HOSSAM
CyS***



EmpowerWork

**EmpowerWork: Bridging the
Employment Gap for People with
Disabilities**



EMPOWER



WORK



ASSIST



CONNECT

EmpowerWork

Types of disabilities



2.5%



5%



4.5%



3.5%



0.7%

According to World Health Organization (WHO)

Approximately 1.3 billion people worldwide 16%
of population

"No Significant Disability" : 84%

Feature Comparison

abilityJOBS

- ✓ Job Matching
- ✗ Less useful for Middle East users



- ✓ Remote Job Opportunities
- ✗ Smaller employer network



- ✓ Employer Partnerships
- ✗ Not Global



- ✓ Accessibility Tools
- ✗ Primarily focused on India



EmpowerWork

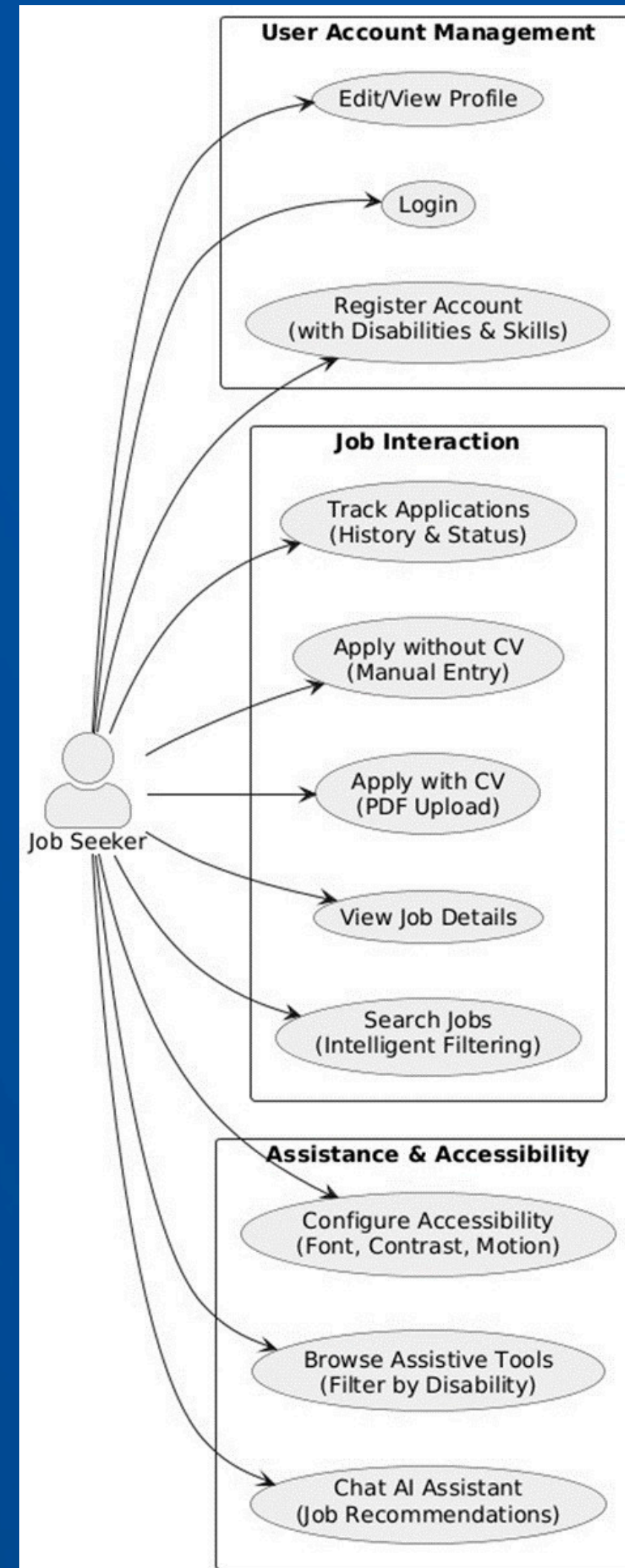
Advantages:

1. Strong Social Impact & Inclusivity
2. Advanced AI-Driven Features
3. Flexible and Future-Ready Design
4. Freelancing

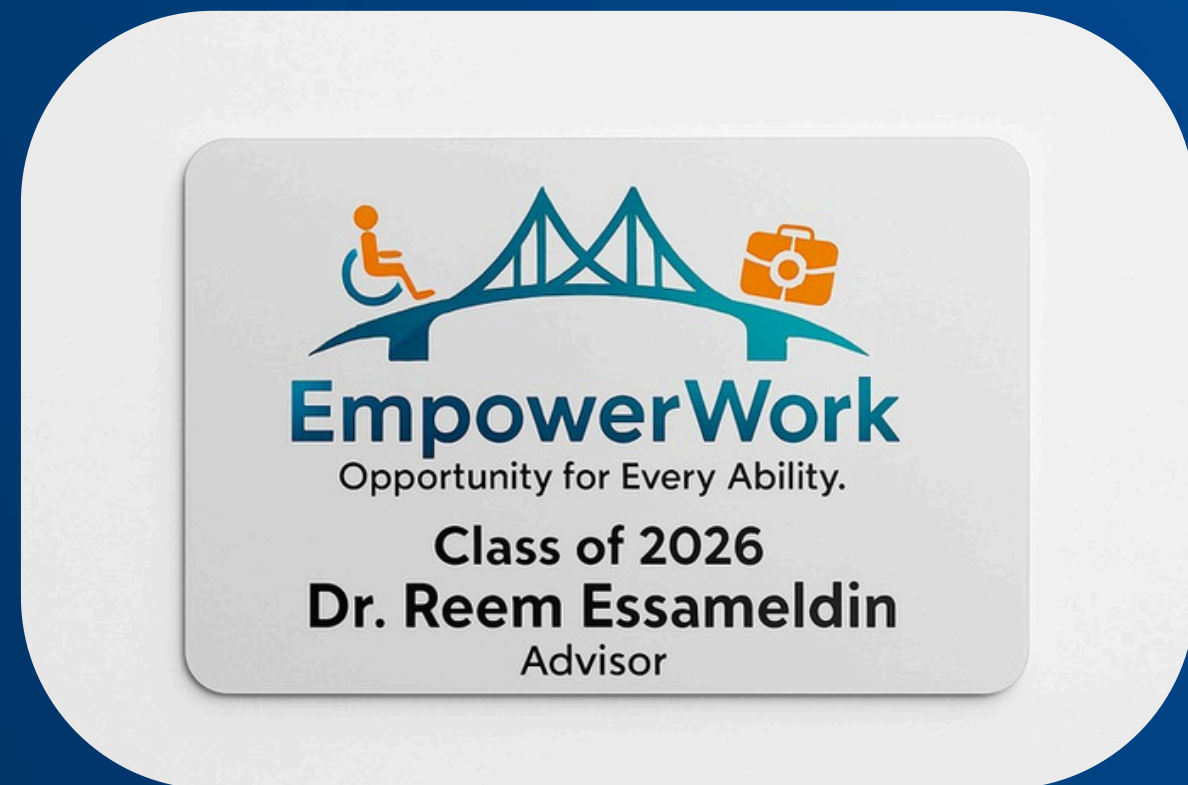


Diagrams

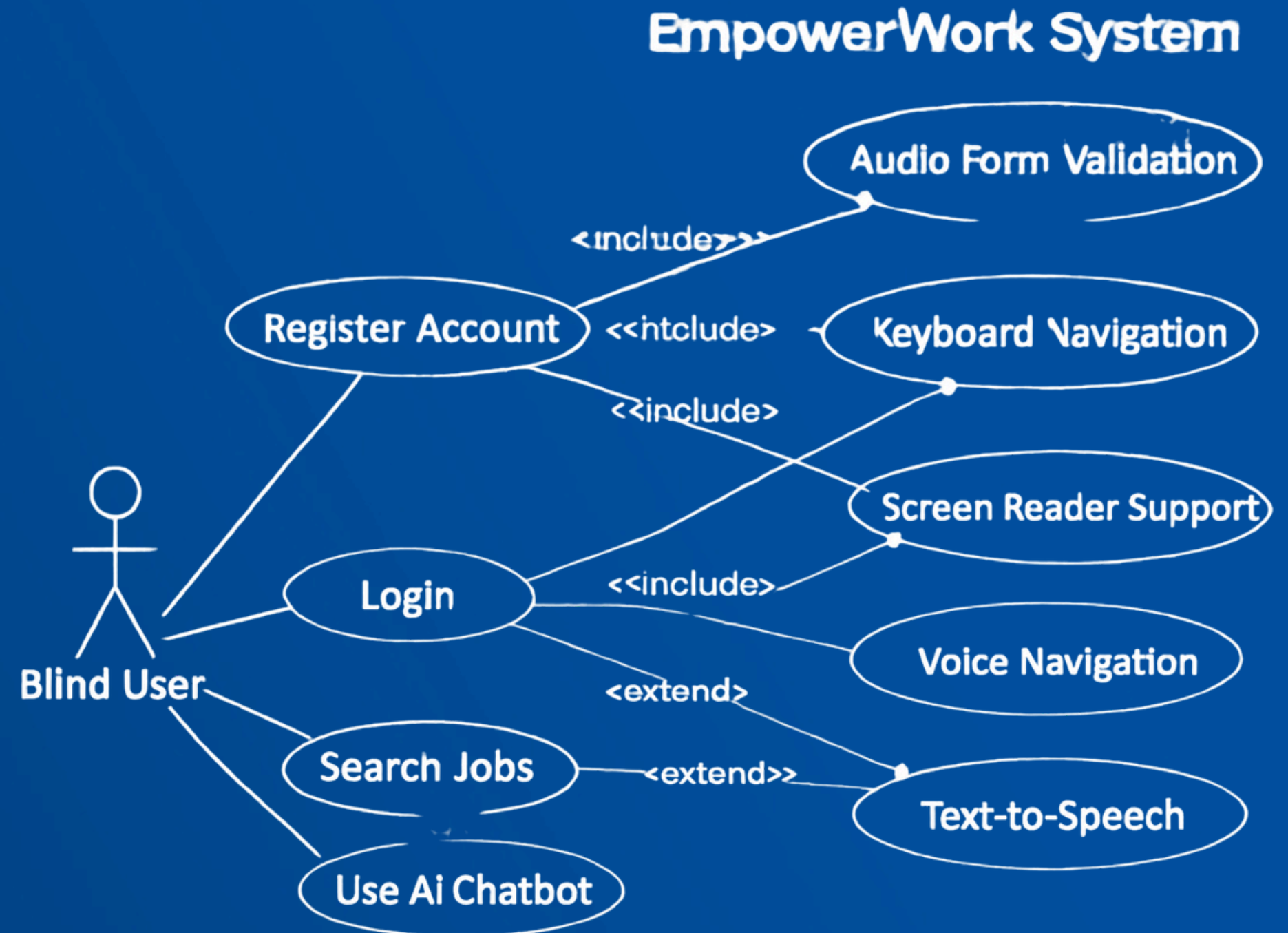
Use case diagram (Normal person)



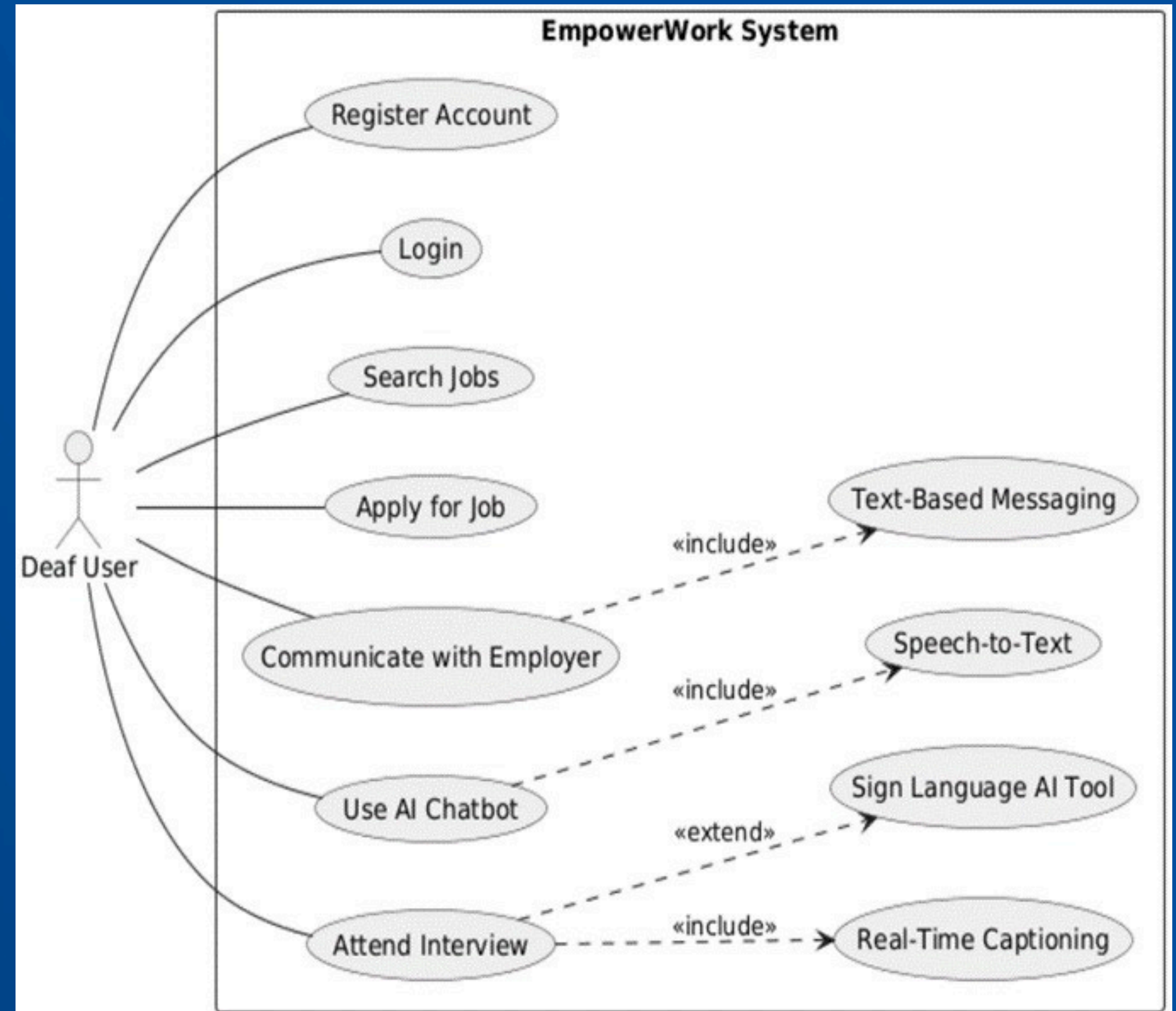
Use case diagram (Blind person)



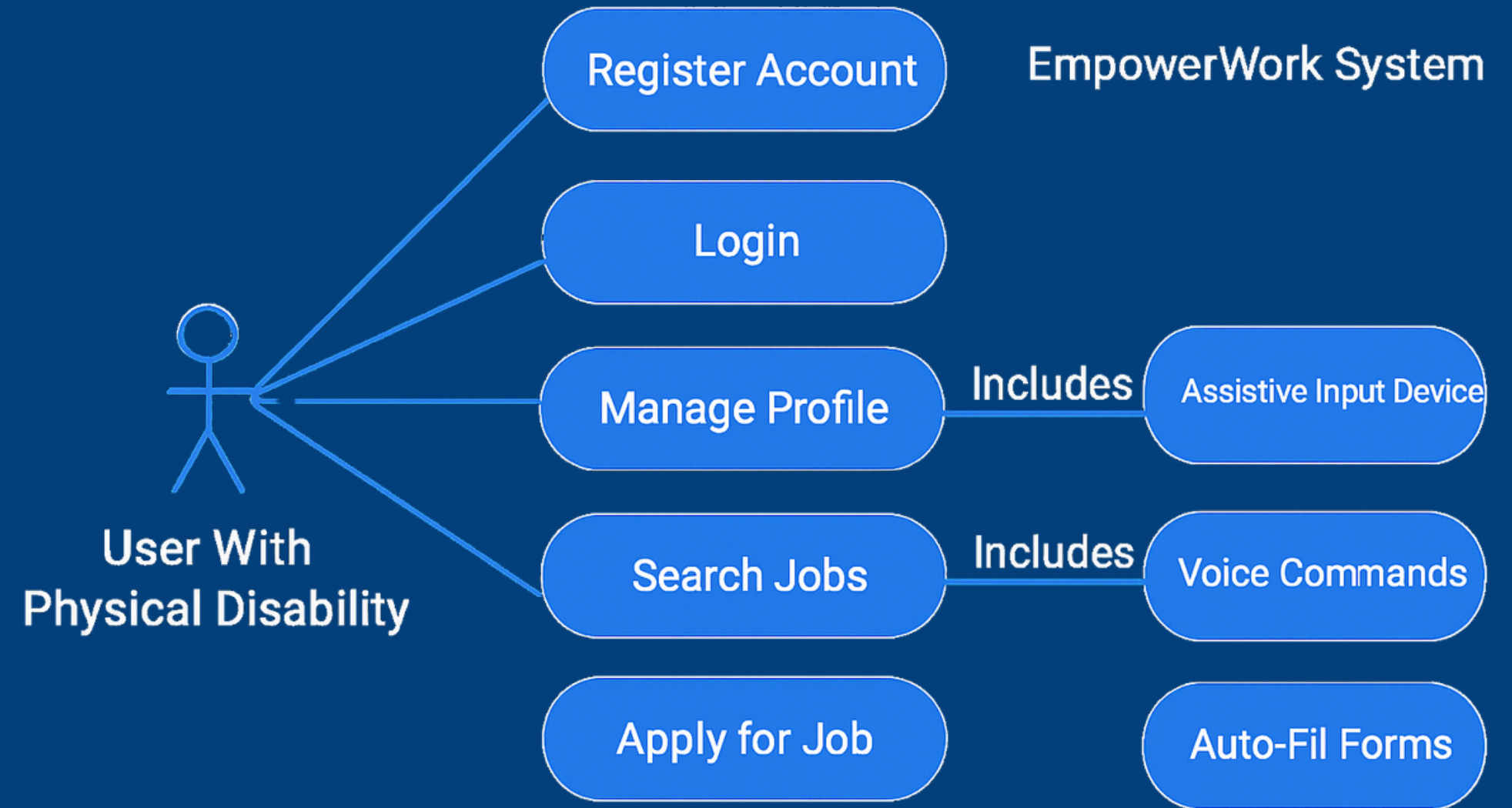
NFC CARD



Use case diagram (Deaf person)



Use case diagram (Physical Disabled Person)



Our offered solutions



Our AI recruitment tools

1. sign language
2. cv generator
- 3 .action recognition



Assistive Communication Technologies

5. speech to text
6. text to speech
7. intrusion detection

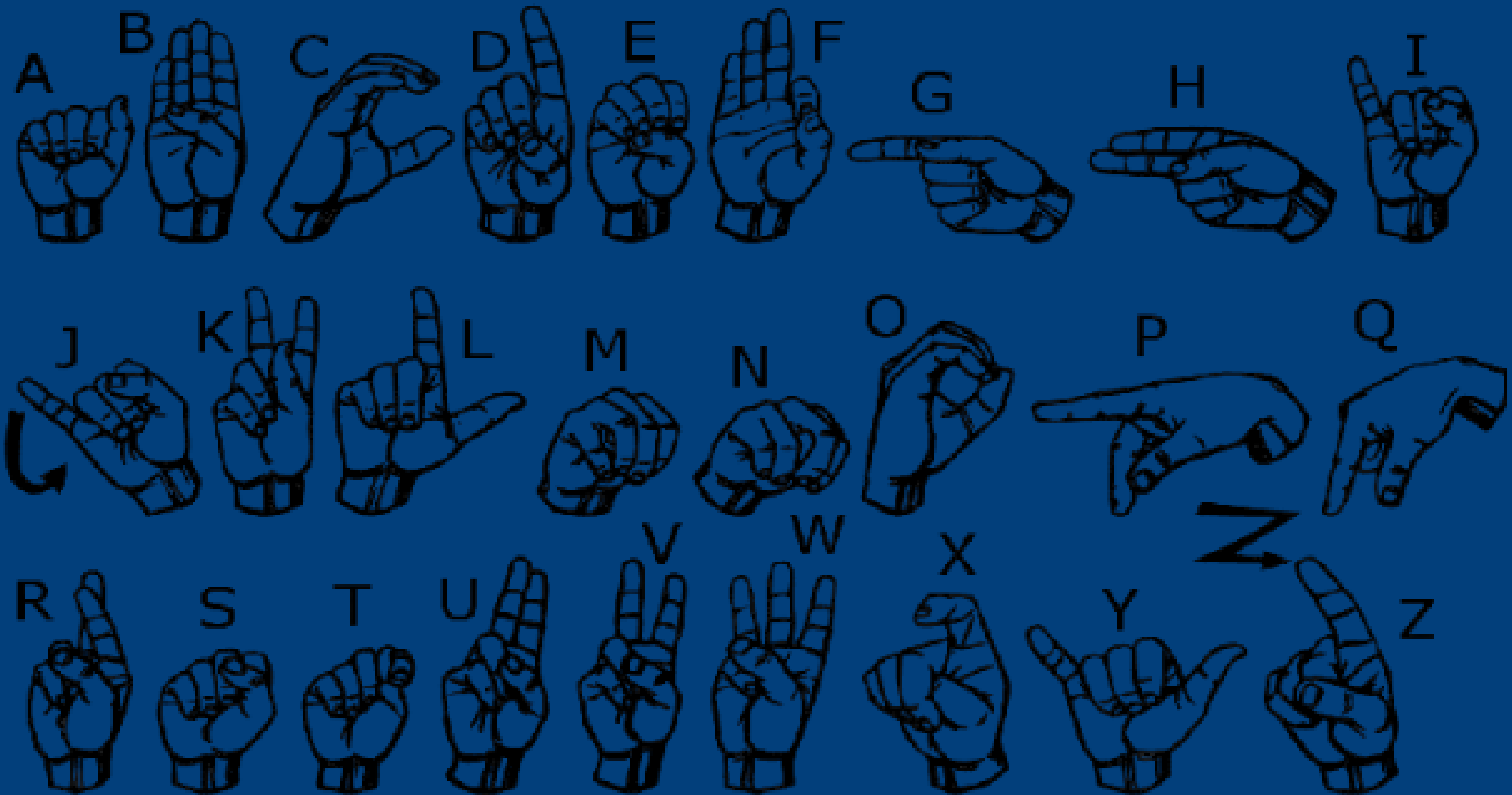
SIGN LANGUAGE

Our Solution: A Real-Time Bridge



To develop a system that accurately translates sign language gestures into text and speech, facilitating seamless communication between the hearing-impaired community and the rest of the world.

Data Sample



Landmark Extraction

- Use **MediaPipe Hands**
- MediaPipe converts each RGB frame into **21 hand landmarks**
- Each landmark has **(x, y, z)** coordinates
- Provides accurate skeletal representation of the hand



98.98% of samples were classified correctly.				
	precision	recall	f1-score	
A	1.00	0.99	0.99	
B	1.00	0.99	1.00	
C	1.00	1.00	1.00	
D	0.99	0.99	0.99	
E	0.99	1.00	0.99	
F	1.00	1.00	1.00	
G	0.99	0.99	0.99	
H	0.99	0.99	0.99	
I	1.00	0.98	0.99	
J	0.99	0.99	0.99	
K	1.00	1.00	1.00	
L	1.00	1.00	1.00	
M	0.92	0.96	0.94	
N	0.96	0.90	0.93	
O	0.99	0.99	0.99	
P	1.00	0.99	0.99	
Q	1.00	1.00	1.00	
R	0.99	0.99	0.99	
S	0.98	0.99	0.99	
T	0.99	0.99	0.99	
U	0.99	0.98	0.99	
V	1.00	0.99	1.00	
W	0.99	0.99	0.99	
X	0.99	0.99	0.99	
Y	1.00	1.00	1.00	
Z	0.99	0.99	0.99	
del	0.98	0.99	0.99	
space	0.98	1.00	0.99	
accuracy			0.99	
macro avg			0.99	
weighted avg			0.99	

classification

report



🔮 **REAL-TIME GESTURE PREDICTION**

Updates in millisecond real-time based on your hand landmarks.

S

Classified Gesture Name

CONFIDENCE ACCURACY METER

Calculated probabilistic certainty of the ML RandomForest classifier.

Accuracy Status



75%

Moderate Signal; Hold Hand Still

CV GENERATOR

CV GENERATION



AI Voice Interview

All Done! Please review and click Generate.

Profile Completion

100%

Full Name

Rawan Mohamed

Title

developer

Email

rawanmohamedf@gmail.com

Phone

01091221840

Experience

studied at anu

Education

graduated from anu

Skills

developer



Download Professional PDF Resume



AI Voice Interview

Step 1 of 7

AI is speaking: Asking for name...

Profile Completion

0%

Full Name

Waiting for name...

Title

Waiting for title...

Email

Waiting for email...

Phone

Waiting for phone...

Experience

Your experience will appear here...

Mohammed Gamal

AI Engineer

Mohammed@gimil.com | 01276351034

Summary

Highly motivated and experienced AI Engineer with a strong background in computer science and data analysis, possessing a unique blend of technical skills and collaborative mindset to drive innovative solutions and deliver high-quality results in fast-paced environments.

Experience

As a dedicated and results-driven professional I have had the opportunity to be a member of the esteemed Creators of Empower Work where I have been able to apply my technical expertise and passion for artificial intelligence to contribute to the development of cutting-edge projects and solutions that drive business growth and improvement.

Education

I have had the privilege of being educated at the prestigious Faculty of Computer and Data Science where I have gained a solid foundation in computer science and data analysis, and have developed a deep understanding of the theoretical and practical aspects of artificial intelligence and its applications.

Skills

I possess excellent skills that enable me to work effectively under pressure and thrive in team-based environments, with the ability to collaborate with cross-functional teams to achieve common goals and deliver high-quality results in a timely and efficient manner.

Contact

To discuss my qualifications and experience in more detail please do not hesitate to contact me at Mohammed@gimil.com or 01276351034

Action Recognition

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Action Recognition

Upload a video to analyze activities, objects, and significant moments using AI vision.

Video Input

[Upload File](#)[Live Camera](#)

Click or drag a video here

MP4, WebM, MOV supported

[Extract Actions](#)

Analysis Report



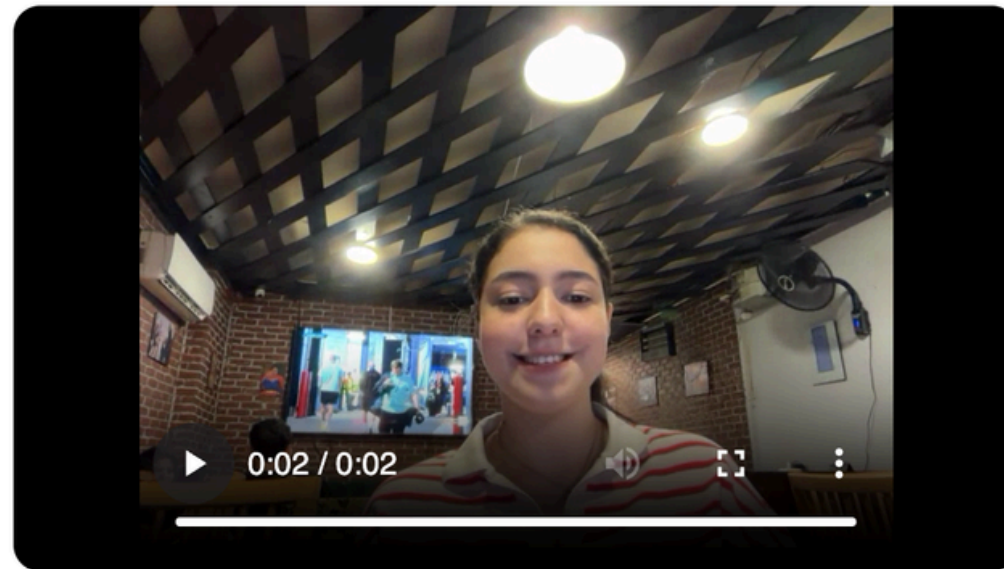
Upload and analyze a video to see the generated intelligence report here.

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Action Recognition

Upload a video to analyze activities, objects, and significant moments using AI vision.

Video Input



live_capture_1782494760917.webm

[Remove](#)

 Extract Actions

Analysis Report



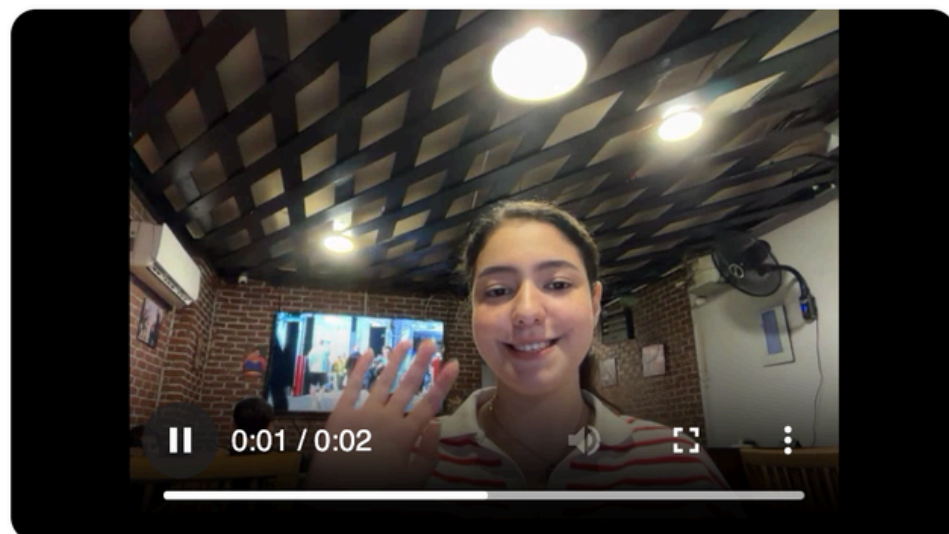
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Action Recognition

Upload a video to analyze activities, objects, and significant moments using AI vision.

Video Input



live_capture_1782494760917.webm

[Remove](#)

Extract Actions

Analysis Report

EXECUTIVE SUMMARY\n=====\nHere is a brief executive summary:

The video shows a young woman interacting with the camera in a casual dining establishment, posing, waving, and smiling over a 2.1-second period. The main subject, wearing a red and white striped polo shirt, is the focal point, while the background setting includes a large TV screen, wooden tables and chairs, and several people engaged in their own activities. The woman's friendly demeanor and gestures suggest a vlogging or social media recording activity in a relaxed, social environment.\n\nDETAILED

REPORT\n=====\n**Video Analysis Report (2.1 seconds)**

1. Overall Summary of Actions and Activities Observed:

The video shows a young woman interacting with the camera in a casual dining establishment, likely a sports bar or restaurant. The woman is the main subject, and her actions include posing for the camera, waving, and smiling. The background setting shows people seated at tables, watching



- Llama-4-Scout-17B Top-1 Accuracy, Generates natural language 86% to 89% accuracy

←

👁️

Action Recognition

Upload a video to analyze activities, objects, and significant moments using AI vision.

📺

Video Input

0:00 / 0:02

8192e642-f024-4def-8ece-e97e9f0f1cb9.webm

Remove

👁️

Extract Actions

✅

Analysis Report

EXECUTIVE SUMMARY\n=====\nHere is a brief executive summary:

The 2.8-second video shows a young woman in a pink polo shirt waving and greeting the camera in a friendly and professional manner. The main activities observed are her waving and direct gaze at the camera, suggesting a welcoming demeanor. The video appears to be a brief introduction or greeting, possibly for a video recording or live stream, set against a decorative white wall with artificial green leaves.\n\nDETAILED REPORT\n=====\n**Video Analysis Report (2.8 seconds)**

1. Overall Summary of Actions and Activities Observed:

The video shows a young woman with dark hair pulled back, wearing a pink or salmon-colored polo shirt, standing in front of a decorative white wall with artificial green leaves. The main actions observed are the woman waving her right hand and appearing to greet or acknowledge the camera. The overall atmosphere seems casual, friendly, and professional.

Output detailed report

✓ Analysis Report

EXECUTIVE SUMMARY\n=====\nHere is a brief executive summary:

The 2.8-second video shows a young woman taking a selfie in a casual indoor setting, likely a restaurant or cafe with a rustic decor. The main activities observed are the woman posing for the camera, making a gesture with her right hand, and changing her facial expression from neutral to a smile. The key participant is the woman, wearing a red and white striped polo shirt, with notable background features including an exposed brick wall and wooden ceiling.\n\nDETAILED REPORT\n=====\n**Video Analysis Report (2.8 seconds)**

1. Overall Summary of Actions and Activities Observed

The video appears to show a young woman taking a selfie in an indoor location. The actions and activities observed are limited, with the woman posing for the camera and possibly making a gesture with her right hand. The overall atmosphere seems casual and relaxed.

2. Key Moments and Notable Events

- * At Frame 0 (0.0s), the woman is seen taking a selfie with a neutral facial expression.
- * At Frame 30 (1.2s), the woman's right hand is visible, and she appears to be making a gesture, possibly waving or adjusting the camera. Her facial expression has changed to a smile.
- * At Frame 60 (2.5s), the woman is seen posing for the camera with a subtle smile, and her facial expression appears more relaxed.

3. Description of Participants/Objects

- * The main participant in the video is a young woman with:
 - + Dark hair pulled back
 - + Wearing a red and white striped polo shirt
 - + Possibly wearing a thin necklace (visible in Frame 60)
- * Background objects and features:
 - + Exposed brick wall
 - + Wooden ceiling with beams (possibly with diamond-shaped patterns)
 - + Lighting fixtures (possibly overhead lighting)
 - + A slanted ceiling (visible in Frame 60)

4. Timeline of Main Activities

- * 0.0s (Frame 0): The woman takes a selfie with a neutral facial expression.
- * 1.2s (Frame 30): The woman makes a gesture with her right hand and smiles.
- * 2.5s (Frame 60): The woman poses for the camera with a subtle smile.

5. Any Patterns or Trends Noticed

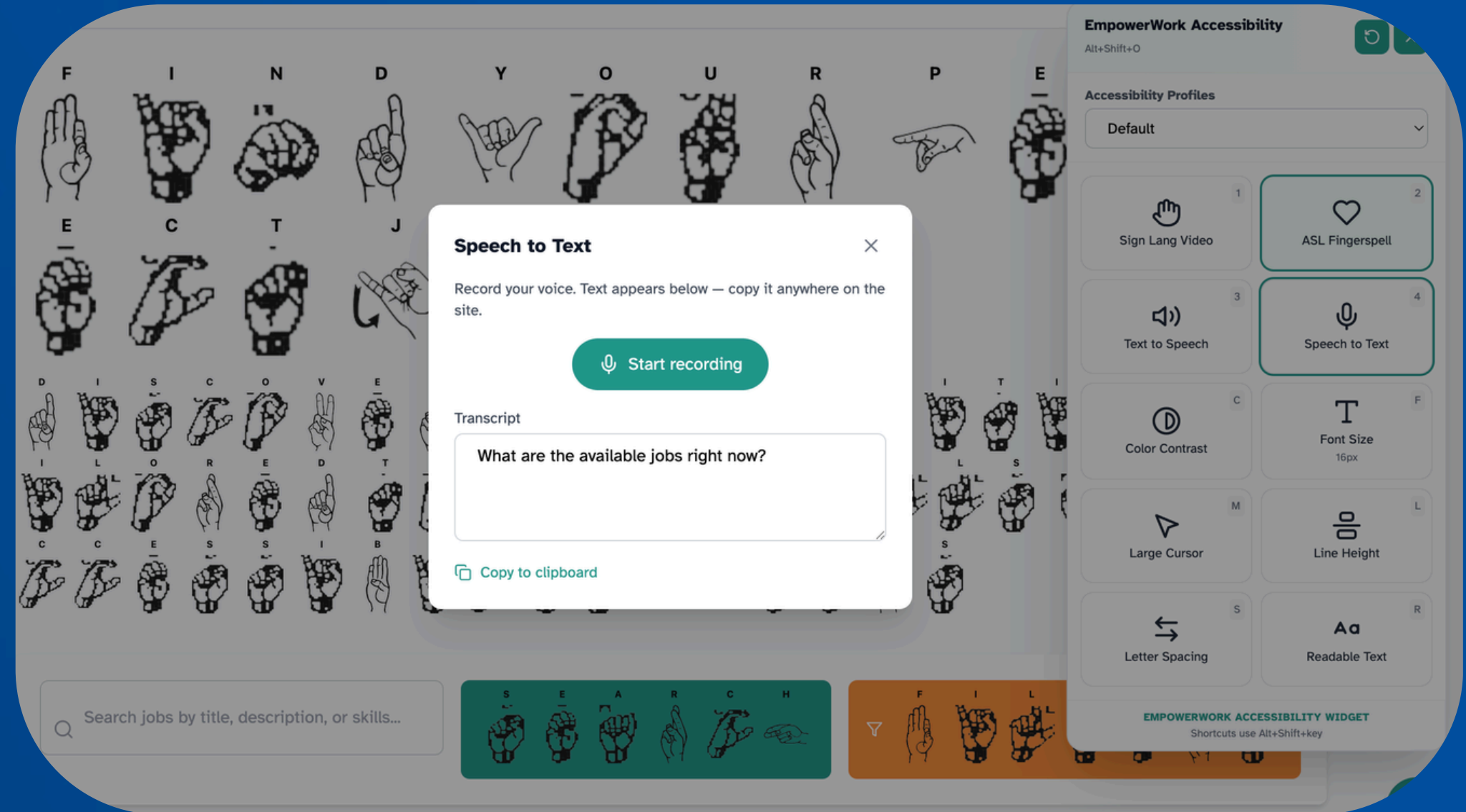
- * The woman's facial expression changes from neutral to a smile over the course of the video.
- * The woman's right hand is only visible in Frame 30, suggesting that she may have been making a gesture or adjusting the camera.
- * The background features suggest a consistent indoor location, possibly a restaurant, cafe, or event space with a rustic or industrial-chic decor.

Overall, the video appears to show a young woman taking a selfie in a casual indoor setting, with limited actions and activities observed. The woman's facial expression and gestures suggest a relaxed and possibly playful atmosphere.

TEXT TO SPEECH & SPEECH TO TEXT

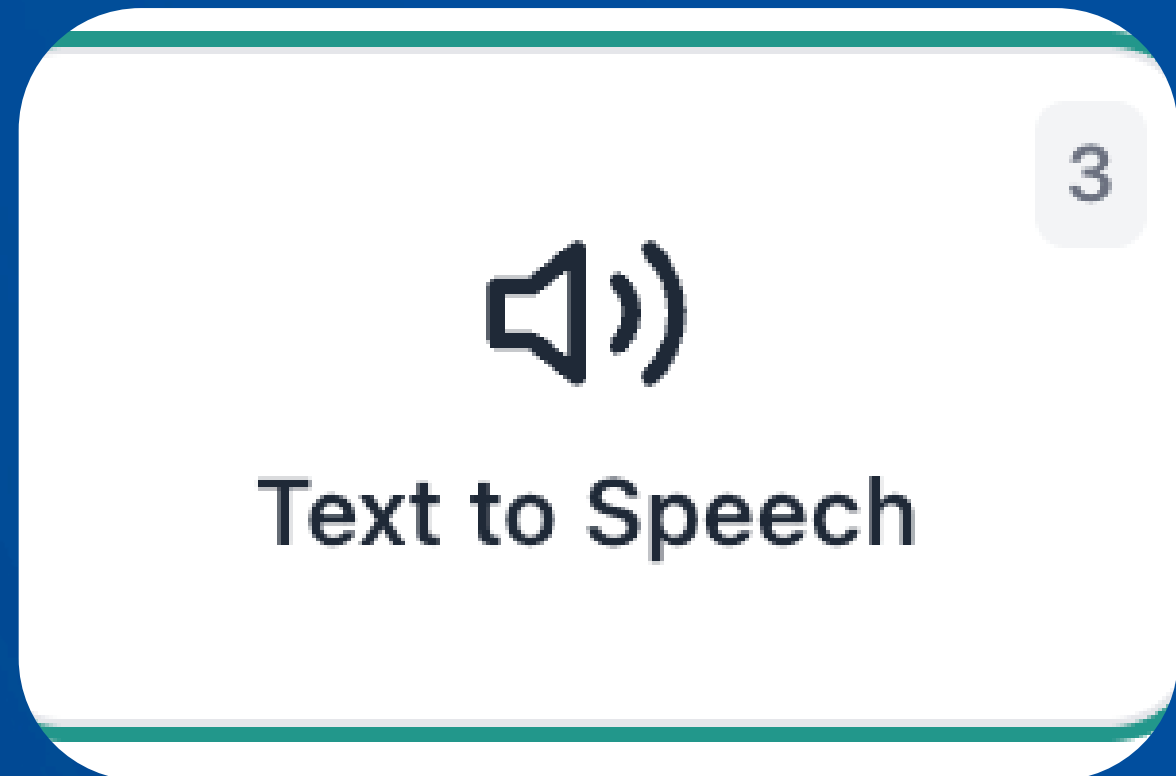
Speech-to-Text (STT):

1. Converts spoken language into text
2. Enables deaf/hard-of-hearing users to read spoken content
3. Supports real-time captioning, voice commands, and note-taking
4. Whisper-Large-v3-Turbo
92.5% – up to 99%



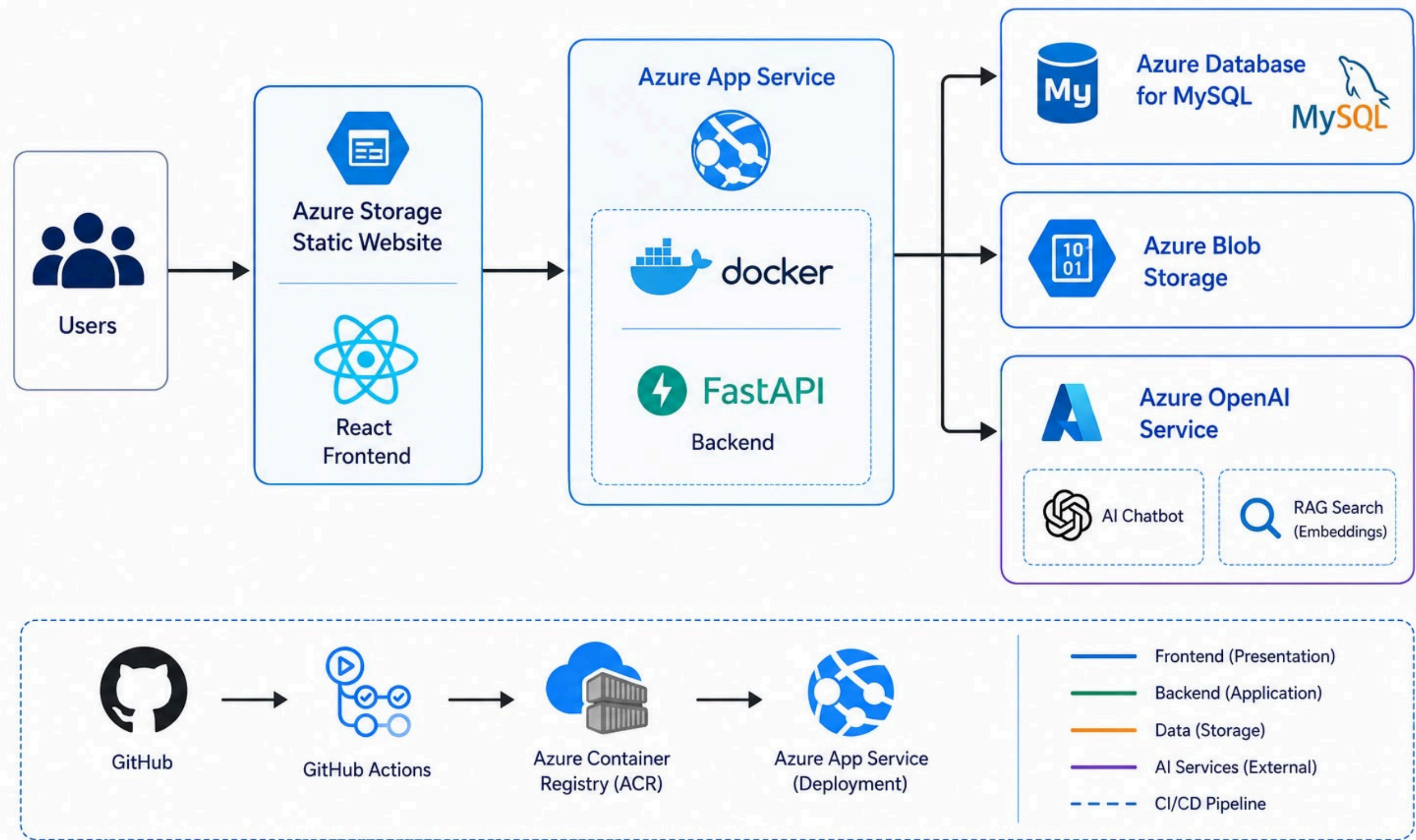
Text-to-Speech (TTS):

1. Converts written text into spoken audio
2. Assists visually impaired users in accessing written content
3. Enhances multitasking and comprehension
4. Web Speech API 97.5% accuracy
0ms network latency, 0\$ server costs,
100% client privacy



AZURE PIPELINE

EMPOWERWORK – AZURE CLOUD ARCHITECTURE



CYBER SECURITY



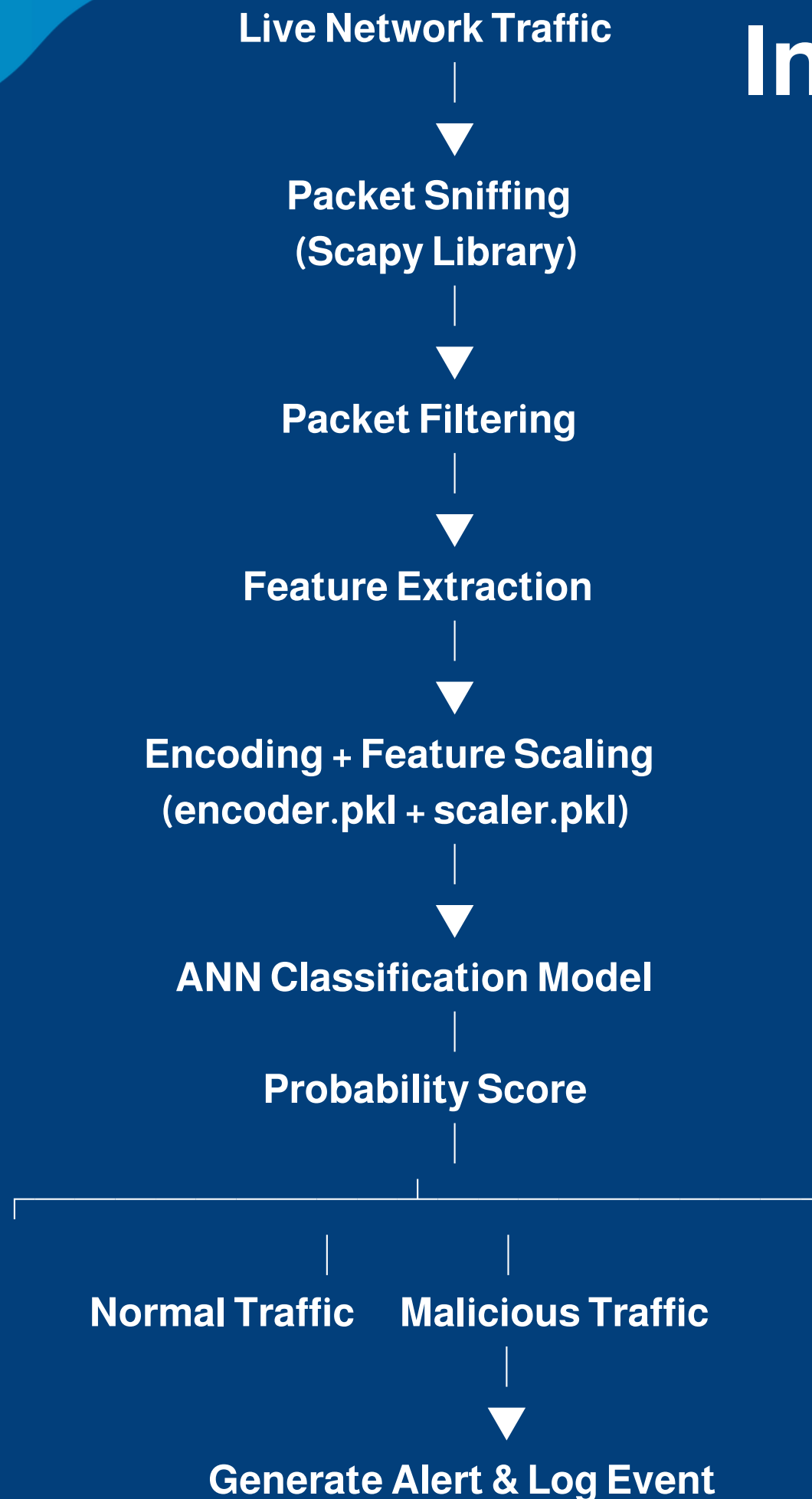
Cybersecurity Goals

- Protect users' personal and health-related information.
- Preserve user privacy and dignity.
- Prevent unauthorized access, data leaks, and harassment.
- Build trust so users can confidently use our AI-powered services



Security is integrated throughout the system, not added afterwards.

Intrusion Detection System (IDS)



```
Sniffing live packets for attack detection...
[2025-11-19 18:45:22.726033] Ether / IP / TCP 172.28.0.12:8080 >
172.28.0.1:52888 PA / Raw -> Normal
[2025-11-19 18:45:22.826176] Ether / IP / TCP 172.28.0.1:52888 >
172.28.0.12:8080 A -> Normal
[2025-11-19 18:45:22.917790] Ether / IP / TCP 172.28.0.12:8080 >
172.28.0.1:52888 PA / Raw -> Normal
[2025-11-19 18:45:23.012345] Ether / IP / TCP 172.28.0.1:52888 >
172.28.0.12:8080 A -> Normal
[2025-11-19 18:45:23.123456] Ether / IP / TCP 172.28.0.12:8080 >
172.28.0.1:52888 PA / Raw -> Normal
[2025-11-19 18:45:23.234567] Ether / IP / TCP 172.28.0.1:52888 >
172.28.0.12:8080 A -> Normal
...
```

- Why Scapy?
- Why feature extraction?
- Why NSL-KDD?
- Why Encoding?
- Why Scaling?
- Why ANN?
- Why Groq?

FRONTEND

Assistive Communication Technologies

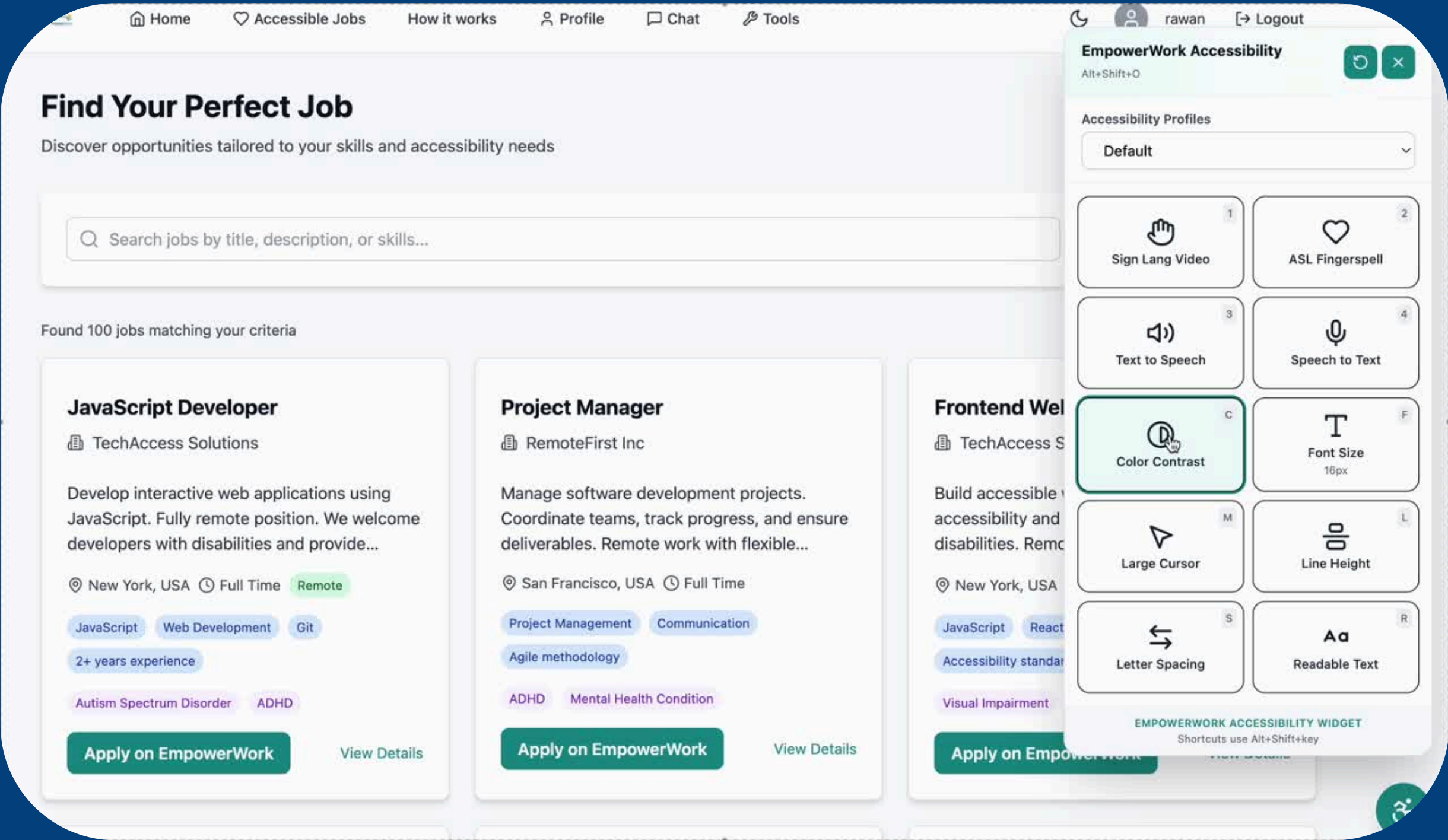
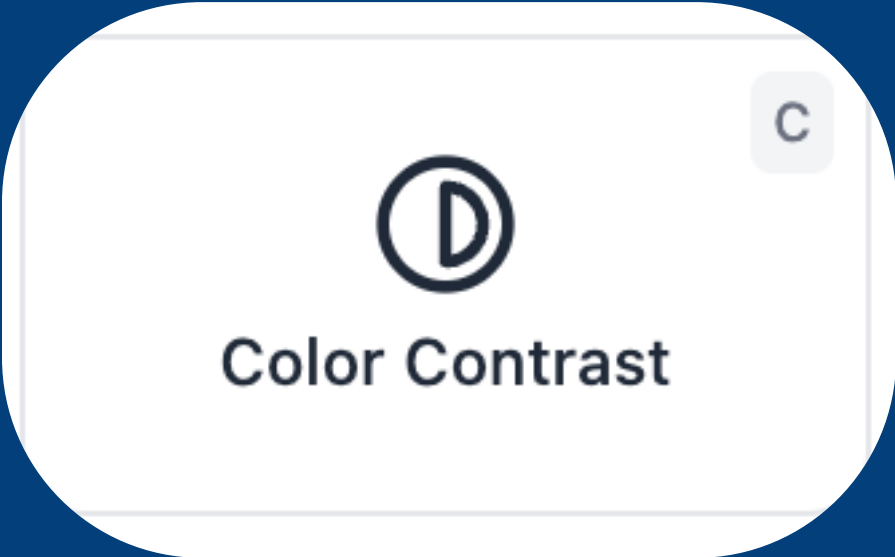
2

ASL Fingerspell

The screenshot displays the EmpowerWork Accessibility website interface. The top navigation bar includes links for Home, Accessible Jobs, How it works, Profile, Chat, and Tools. A user profile for 'rawan' is logged in, with a Logout option. The main heading is 'Find Your Perfect Job', followed by the subtext 'Discover opportunities tailored to your skills and accessibility needs'. A search bar prompts users to 'Search jobs by title, description, or skills...'. Below the search bar, it states 'Found 100 jobs matching your criteria'. Three job listings are visible:

- JavaScript Developer** by TechAccess Solutions: Develop interactive web applications using JavaScript. Fully remote position. We welcome developers with disabilities and provide... Location: New York, USA. Type: Full Time. Remote. Skills: JavaScript, Web Development, Git. Experience: 2+ years experience. Accessibility: Autism Spectrum Disorder, ADHD. Button: Apply on EmpowerWork. View Details.
- Project Manager** by RemoteFirst Inc: Manage software development projects. Coordinate teams, track progress, and ensure deliverables. Remote work with flexible... Location: San Francisco, USA. Type: Full Time. Skills: Project Management, Communication. Methodology: Agile methodology. Accessibility: ADHD, Mental Health Condition. Button: Apply on EmpowerWork. View Details.
- Frontend Web Developer** by TechAccess Solutions: Build accessible web applications with a focus on accessibility and disabilities. Remote. Location: New York, USA. Skills: JavaScript, React. Standards: Accessibility standards. Accessibility: Visual Impairment. Button: Apply on EmpowerWork. View Details.

On the right side, the 'EmpowerWork Accessibility' widget is open, showing various accessibility profiles and settings. The 'Default' profile is selected. The widget includes icons and labels for: Sign Lang Video, ASL Fingerspell, Text to Speech, Speech to Text, Color Contrast, Font Size (16px), Large Cursor, Line Height, Letter Spacing, and Readable Text. At the bottom of the widget, it states 'EMPOWERWORK ACCESSIBILITY WIDGET' and 'Shortcuts use Alt+Shift+key'.



T

Font Size

16px

F

Home

Accessible Jobs

How it works

Profile

Chat

Tools

rawan

Logout

Find Your Perfect Job

Discover opportunities tailored to your skills and accessibility needs

Search jobs by title, description, or skills...

Found 100 jobs matching your criteria

JavaScript Developer

TechAccess Solutions

Develop interactive web applications using JavaScript. Fully remote position. We welcome developers with disabilities and provide...

New York, USA

Full Time

Remote

JavaScript

Web Development

Git

2+ years experience

Autism Spectrum Disorder

ADHD

Apply on EmpowerWork

View Details

Project Manager

RemoteFirst Inc

Manage software development projects. Coordinate teams, track progress, and ensure deliverables. Remote work with flexible...

San Francisco, USA

Full Time

Project Management

Communication

Agile methodology

ADHD

Mental Health Condition

Apply on EmpowerWork

View Details

Frontend Web Developer

TechAccess Solutions

Build accessible web applications with a focus on accessibility and disabilities. Remote work with flexible...

New York, USA

JavaScript

React

Accessibility standards

Visual Impairment

Apply on EmpowerWork

View Details

EmpowerWork Accessibility

Alt+Shift+O

Accessibility Profiles

Default

1

Sign Lang Video

2

ASL Fingerspell

3

Text to Speech

4

Speech to Text

C

Color Contrast

F

Font Size

16px

M

Large Cursor

L

Line Height

S

Letter Spacing

R

Aa

Readable Text

EMPOWERWORK ACCESSIBILITY WIDGET

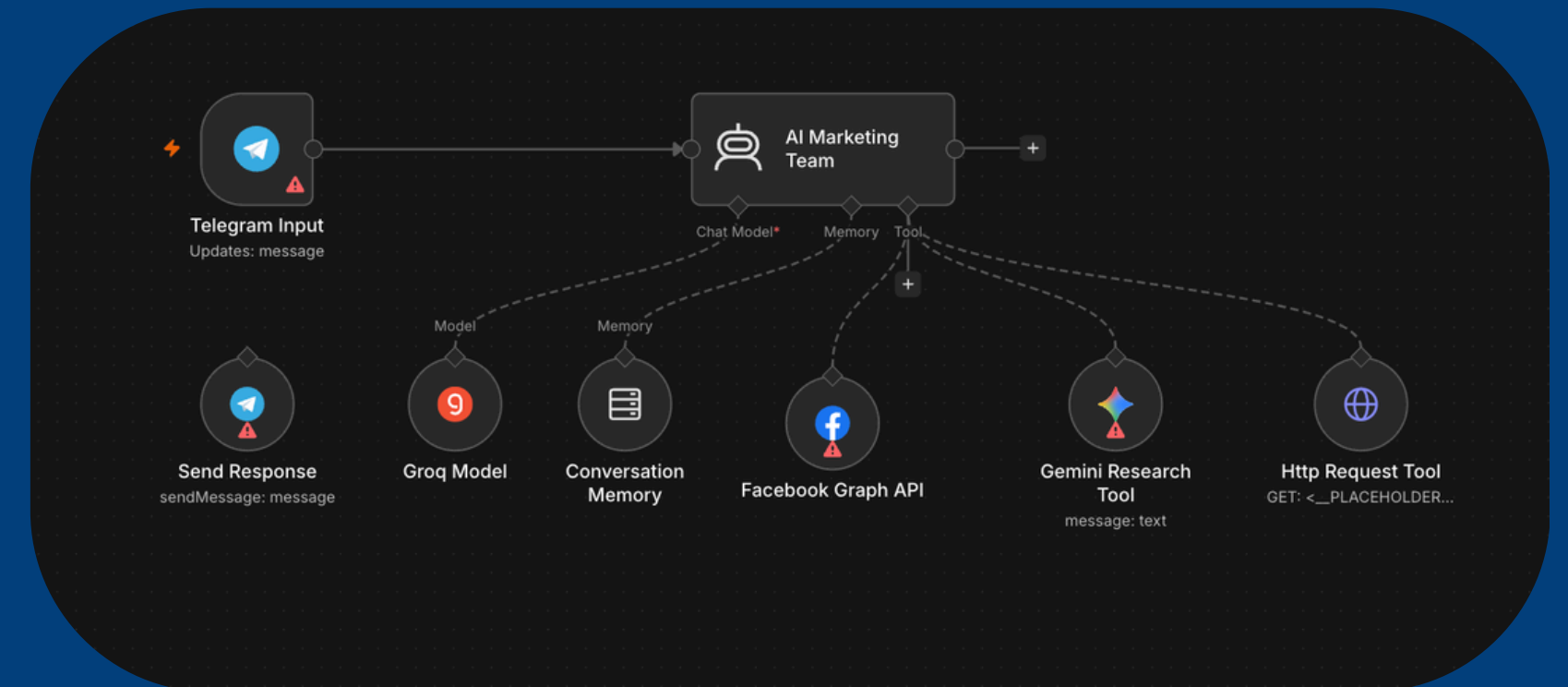
Shortcuts use Alt+Shift+key

FUTURE PLANS

1. CONTRACT WITH WAZZUF ,
LINKEDIN AND FORASNA

2. AI AUTOMATION N8N FOR
MARKETING

3. BOT USES ALL FEATURES
AUTOMATICALLY



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Find Your Perfect Job

Discover opportunities tailored to your skills and accessibility needs

[Search](#)[Filters](#)

Found 8 jobs matching your criteria

JavaScript Developer

TechAccess Solutions

Develop interactive web applications using JavaScript. Fully remote position. We welcome developers with disabilities and provide...

New York, USA Full Time Remote

JavaScript Web Development Git

2+ years experience

[Apply](#)[View Details](#)

Project Manager

RemoteFirst Inc

Manage software development projects. Coordinate teams, track progress, and ensure deliverables. Remote work with flexible...

San Francisco, USA Full Time

Project Management Communication

Agile methodology

[Apply](#)[View Details](#)

Graphic Designer

Diverse Talent Hub

Create visual designs for digital and print media. Remote position with flexible hours. We provide design software and tools that supp...

Sydney, Australia Part Time Remote

Graphic Design Adobe Creative Suite Creativity

[Apply](#)[View Details](#)

Frontend Web Developer

Content Writer

Customer Service Representative

WAIT BEFORE THANK YOU

Supervised by

Dr. Reem Essameldin

Rawan Mohamed Farouk	2205082	Intelligent Systems
Khaled Ghalwash	2205018	Intelligent Systems
Mohamed Gamal	2205131	Intelligent Systems
Mohamed Hassan	2205212	Intelligent Systems
Nadeen Ehab	2205081	Cyber Security
Mazen Hossam	2205041	Cyber Security